National University of Computer and Emerging Sciences
School of Computing
Islamabad Campus
AI2002: Artificial Intelligence
Semester Project - Fall 2026

PROJECT STATEMENT

Objective

Each group is required to design, implement, test, and demonstrate a complete project that addresses a clearly defined real-world problem and applies Artificial Intelligence concepts in a meaningful manner.

The project idea may be selected by the students; however, the implementation MUST comply with exactly one of the two project paths defined in this statement.

A project that does not satisfy the mandatory requirements of its selected path will be considered non-compliant, regardless of its complexity or quality in other areas.

Project Path Selection

Each group MUST select exactly ONE project path before implementation begins:

Path A: Software-Based

Path B: Hardware-Based

A group MUST NOT submit a project as a combination of Path A and Path B. The selected path determines the mandatory technical requirements and evaluation of the project.

Note: Each group must consist of exactly 2-3 members. Individual projects are NOT allowed.

Path A: Software-Based Project

- A Path A project MUST be a software-based AI application whose core functionality depends on a **Machine Learning algorithm/Neural Network.**
- The project MUST NOT be merely a conventional software application, CRUD system, website, dashboard, or rule-based program with an AI-related label.
- At least ONE Machine Learning algorithm/Neural Network MUST be implemented and used to produce a meaningful **prediction, classification, regression, clustering, recommendation, detection, or other learning-based outcome**.
- The chosen Machine Learning algorithm MUST be justified against a **specific real-world use case**. The group MUST explain why the selected algorithm is appropriate for the problem, what input data it requires, what output it produces, and how the output supports the intended real-world decision or task.
- Generic statements such as 'AI is useful in this field' or 'Machine Learning makes the system intelligent' will NOT be accepted as justification.
- A suitable dataset MUST be identified and documented. The group MUST explain the source, relevant features, target/output (where applicable), preprocessing, and data limitations.

- The Machine Learning pipeline MUST be implemented and evaluated using appropriate **performance metrics**. Results MUST be reported and interpreted.
- **Streamlit deployment** is **MANDATORY**. The final submission MUST include a working Streamlit application through which a user can interact with the implemented ML solution.
- A notebook-only implementation, command-line-only implementation, local script without Streamlit deployment, or static screenshots of an application will NOT satisfy the Path A requirement.
- The Streamlit application MUST demonstrate the complete usable flow: input/data entry, model execution, and understandable presentation of the resulting prediction/analysis.

Path B: Hardware-Based Project

- A Path B project MUST be centered on a **physical hardware component or system** and MUST address a genuine **real-world problem**.
- Any appropriate algorithm may be used, including foundational Artificial Intelligence and Machine Learning algorithms/Neural Network covered in the course, provided that the algorithm contributes directly to solving the stated problem.
- The problem MUST be practically justified. The group MUST identify the **real-world user, situation, limitation, or need being addressed and explain why the proposed system is useful**.
- Hypothetical, trivial, or purely demonstrative hardware projects without a meaningful real-world application will NOT be accepted.
- Where applicable, the project MUST explicitly demonstrate hardware-software(AI-Algorithm) integration, including relevant **sensors, actuators, microcontrollers, embedded systems, communication modules, or other hardware components**.
- The group MUST clearly explain the role of each **hardware component, the data collected or action performed, the algorithmic processing, and the final system response**.
- The system MUST be demonstrated as an integrated working prototype.
- If a software interface is used to monitor or control the hardware, that interface MUST be **functional and clearly connected to the physical system**.

Scope and Project Requirements

- The project problem MUST be **specific**, **measurable**, and **sufficiently substantial** for a semester project.
- The proposed solution MUST contain a clearly identifiable AI/ML component. Merely adding an AI label to a non-AI system is not acceptable.

- The group MUST define the problem, users/stakeholders, inputs, processing/algorithm, outputs, and expected impact before implementation.
- The project MUST be implemented by the group. Directly copying an existing project, tutorial, repository, or complete solution is strictly prohibited.
- External datasets, libraries, APIs, pretrained models, simulators, and hardware modules may be used where appropriate, but the group MUST **understand**, **document**, and **justify their use**.
- The project MUST be tested using **representative inputs** and MUST include evidence that the implemented system works as intended.
- The final implementation MUST be sufficiently complete to support **a live demonstration**. Partially implemented prototypes will be evaluated according to the functionality actually demonstrated.

Mandatory Deliverables

Each **group** MUST submit all of the following ***(only one submission per group)***:

1. **Source Code:** Complete, organized, and runnable source code. The code MUST include appropriate comments, meaningful naming, and a clear project structure.
2. **Project Report:** The report MUST contain the problem statement, real-world justification, selected path, objectives, methodology, algorithm description, system architecture/workflow (a proper pipeline diagram), dataset or hardware description, implementation details, testing, results, limitations, and conclusion.
3. **Demonstration:** A live demonstration MUST be provided. **Path A** groups MUST demonstrate the working Streamlit application. **Path B** groups MUST demonstrate the working hardware-software prototype.
4. **Presentation Slides:** Slides MUST cover the problem, motivation, proposed solution, selected AI/ML technique, implementation details, testing details, results (both qualitative and quantitative), limitations, and conclusion.
5. **Video:** Path B groups MUST Submit Video Link of proper working of hardware. **Note:** Upload the video on google drive and submit only the link of the video.
5. **Evidence of Evaluation: Path A** groups MUST provide relevant ML evaluation results and metrics. Path B groups MUST provide appropriate evidence of system testing (in the form of pre-recorded video) and algorithmic performance. **Note:** Live demonstration is mandatory even if you have shown pre-recorded video. pre-recorded video is not the substitute for a live demonstration.

Timeline

All deadlines are **FINAL**. Late submission, incomplete submission, or failure to demonstrate the required components may result in deduction of marks or rejection of the relevant deliverable.

Action	Deadline
Finalizing Group + path trajectory(software/hardware)	11 Oct 2026 (Sunday) **Submit on:** https://bit.ly/4hEnFnA
Finalizing Dataset + Models + Use Case + Hardware	25 Oct 2026 (Sunday) **Submit on:** https://bit.ly/4h3WC54
Final Project Submission with all required deliverables	06 Dec 2026 (Sunday) **Submit on GCR**
Final Presentations/Demonstrations	07-09 Dec 2026 (Mon-Wed) **Submit on GCR**

Note: Each group must have a **unique project idea**. No two groups are allowed to work on the same idea.

Project ideas will be allocated on a **first-come, first-served basis**. Therefore, make sure to:

- Add your project idea to the **provided sheet** as soon as possible.
- Check the sheet before selecting an idea to ensure it has not already been taken.
- **Lock your cell** after entering your group details so that no other group can overwrite or modify your entry.

Once an idea has been added and the cell is locked, it will be considered **reserved for that group**.

Submission and Academic Integrity

- All required files MUST be zipped together and submitted on GCR only (a single zip file per group). Work submitted through any other medium may NOT be considered for grading.
- Only one member per group should submit the final group submission.
- Every group member is responsible for understanding the submitted work **equally** and MUST be able to explain the implementation during the presentation or demonstration.
- Plagiarism, unauthorized copying, submission of another group's work, or submission of a substantially copied online project is strictly prohibited. Any confirmed plagiarism or academic dishonesty will be dealt with according to the applicable university/course policy.

- Use of external resources is permitted for learning and implementation support, but the submitted work MUST represent the group's own understanding and implementation.

Important Compliance Rules

- A project will NOT be considered complete merely because it has a working interface or a sophisticated idea.
- Path A without a Machine Learning algorithm/Neural Network is NOT acceptable.
- Path A without a working Streamlit deployment is NOT acceptable.
- Path B without a genuine real-world problem is NOT acceptable.
- Path B without the required physical hardware component/integration is NOT acceptable.
- A project presented as a mixture of Path A and Path B as a separate third category is NOT permitted.
- Failure to satisfy a mandatory requirement may lead to substantial loss of marks or rejection of the project, even if other components are functional.

Final Expectations

- The project MUST demonstrate that the group can move from a real-world problem to a justified AI-based solution, implement the required technical components, evaluate the solution, and communicate the results professionally.
- Students are expected to prioritize correctness, meaningful application, technical understanding, and demonstrable functionality over unnecessary complexity.

CLO to Project Component Mapping

Project Component	CLO 1	CLO 2	CLO 3
Real-world problem identification & justification	✓		✓
AI/ML algorithm selection & justification	✓	✓	✓
Algorithm implementation		✓	✓
Dataset / hardware integration		✓	✓
System architecture / pipeline		✓	✓
Testing & performance evaluation	✓	✓	✓
Working demonstration	✓	✓	✓
Presentation & technical explanation	✓	✓	✓